

Towards a Constrained Understanding of Europa's Subsurface Driven By Magnetic Induction Sounding

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1 Introduction

1.1 Background

Europa (radius $R_E = 1,560.8$ km) orbits Jupiter at a distance of $9.38R_J$ (Jupiter's radius $R_J \approx 71,492$ km), placing the moon deep within the giant planet's dynamic magnetosphere. Europa's nearly circular, tidally-locked and slightly eccentric ($\approx 1\%$) orbit is approximately contained to the Jovigraphic equatorial plane, orbiting once in about 85.3 hr. Jupiter, a fast rotator, does so in under 10 hr. In consequence, the magnetospheric field continuously sweeps over the moon. Io, the volcanically active Galilean satellite inward of Europa, ejects copious amounts of material (~ 1 ton/s; see Bagenal & Dols, 2020) which is partially ionized (Clark et al., 2020) as it transports radially outward (Devinat et al., 2024). This process populates the inner-middle magnetosphere with a substantial contingent of thermal plasma (Bagenal & Delamere, 2011). Force balance distributes the plasma in a disc-like sheet, forming a Gaussian distribution about the magnetic equator (Phipps & Bagenal, 2021). This thermal plasma couples to the Jovian magnetospheric field lines and corotates with Jupiter, overtaking and bombarding Europa.

Europa possesses a dilute atmosphere generated primarily via sputtering and radiolysis stemming from charged particle irradiation (Vorbürger & Wurz, 2018) as well as sublimation on the sunlit side (Spencer et al., 1999; Roth, 2021). The neutral envelope is composed mostly of O_2 , with H_2O , H_2 , and many persistent trace species (e.g., Saur et al., 1998; Volwerk et al., 2001; Roth et al., 2016; Plainaki et al., 2018; Roth, 2021). Additionally, the existence of plumes of subsurface ocean material venting into ambient space has been (often indirectly) detected (Roth et al., 2014; Jia et al., 2018; Arnold et al., 2019). Such putative plumes are of great interest because they would provide a valuable window into the complex subsurface chemical composition of the moon and internal geophysical structure. As the magnetospheric plasma flows over Europa, electron impact ionization produces a population of cold ionospheric ions, mass-loading the system (Huang & Hill, 1989) and draining momentum from the convecting plasma and magnetospheric field in order to provide the requisite acceleration to the newly created (Europa-genic) ionospheric plasma (Hill & Michel, 1976; Hill, 1979; M. G. Kivelson & Russell, 1983). This causes the magnetospheric field lines to locally bend (i.e., drape) around Europa's upstream hemisphere and pile-up at this location (Saur et al., 1998; Kivelson et al., 2004). As a result, the magnetospheric plasma is deflected around the body of Europa, creating a wake region flowing towards downstream (Szalay et al., 2017). Away from the ionosphere, which possesses marked non-uniformities due to the Hall effect and convective electric field (e.g., Simon et al., 2011, 2021), these electromagnetic perturbations smoothly connect to a system of cylindrical current-carrying wave tubes, Alfvén wings, that extend all the way to Jupiter's polar ionosphere (Neubauer, 1980, 1998; Hess et al., 2011).

Jupiter's magnetic axis is tilted by 9.6° against its polar axis. Thus, at the planetary synodic rotation period $T_s = 11.23$ hr, the icy moon experiences an oscillation in the corotating magnetospheric fields and plasma density. This well-behaved, contin-

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uous undulation can be described in Europa’s reference frame as an ambient field with a constant southward component and a time-varying “horizontal” component (in the equatorial plane). The variability in the horizontal component drives inductive currents in any conducting layers of the satellite system (e.g., a salty subsurface ocean or a dense ionosphere; see Hartkorn et al., 2017). The oscillation in the plasma currents produces only a marginal modification to the inductive response (Schilling et al., 2007), and is typically neglected. For a perfect conducting sphere, the inductive response reduces to a dipole field pointing anti-parallel and equal in magnitude to the horizontal inducing component (Zimmer et al., 2000). The Galileo spacecraft was the first to perform targeted flybys of Europa: data it captured revealed the presence of a conductive spherical layer, thought to be a global subsurface ocean (Kivelson et al., 1999, 2000).

Both by observations and geophysical models, Europa is suggested to possess a differentiated internal structure (Anderson et al., 1997; Schubert et al., 2004), radially distributed into a metallic core, a silicate rock mantle, and a salty liquid water ocean ~ 50 – 200 km thick, all under an icy crust (e.g., Anderson et al., 1998; Kivelson et al., 2000; Zimmer et al., 2000). A complex physical and chemical coupling between the moon’s exterior (i.e., the plasma environment; Johnson, 1990; Carlson et al., 1999; Cassidy et al., 2013) and interior (Ligier et al., 2016; Trumbo & Brown, 2023) facilitates the presence of hydrated sulfates and chlorinated salts across Europa’s surface (Trumbo et al., 2020; Mermy et al., 2023). The eccentricity of Europa’s orbit is crucial in that it maintains the liquid phase of the ocean. Tidal forces owing to Jupiter’s gravitation along a non-circular trajectory flex and heat the moon, injecting energy into the subsurface and providing the bulk of the thermal requirements to maintain liquid water at some depth. Typically, the thermal dissipation of gravitational energy would circularize the orbit, but mean-motion resonance with the neighboring Galilean satellites Io and Ganymede preserves Europa’s slightly eccentric orbital path (Walker & Rhoden, 2022). Hence, the source of internal energy is stable over Europa’s geological timescales (Spohn & Schubert, 2003).

1.2 Motivation

With the discovery of Europa’s putative liquid water ocean, the question of how Europa’s outer subsurface is composed and distributed naturally arises. The presence of a large volume of liquid water combined with the rich geochemical system places Europa as a candidate for biological habitability (Vance et al., 2018; Howell & Pappalardo, 2020). Therefore, it is crucial to constrain the depth, thickness, and composition of Europa’s interior ocean, all of which determine the radial conductivity profile. In this context, brines in the ocean are highly conductive compared to the insulating ice in the crust and mantle (Vance et al., 2021). This is the overarching goal of the upcoming Europa Clipper investigation, a Flagship-class NASA Mission that will perform dozens of grazing flybys of the moon (closest approach $\mathcal{O}(10)$ km; see Cangahuala et al., 2025).

Previous modeling efforts have provided loose bounds for the location, thickness, and conductivity of the ocean. Treating the ambient field as spatially uniform on length scales of several R_E and assuming the subsurface ocean layer to be spherically symmetric, the induced field can be represented by a time-varying dipole moment, centered at Europa. Hence, in practice, a generic dipole field expression is matched to the inductive response with parameters $A \in [0, 1]$ representing the amplitude and $\phi \in [0^\circ, 90^\circ]$ (Zimmer et al., 2000). The values of the parameters then constrain the actual conductivity profile (ocean depth, thickness, and conductivity) that would result in such a signal (Kivelson et al., 2000; Zimmer et al., 2000; Seufert et al., 2011). It remains challenging to identify a “pristine” induction signal from any in-situ observation since the draping and Alfvén wing system produce measurable distortions to the electromagnetic fields near the moon (e.g., Saur et al., 1998) and even modify the signature itself (Neubauer, 1998). The analysis of Zimmer et al. (2000) hinges on this result, as their approach evaluates this expression for varying values of A and ϕ and compares it to the data collected by Galileo along

the E4 and E14 encounters. Their work provides the bounds $A \in [0.7, 1]$ and $\phi \sim 0^\circ$. However, these authors used only a comparison between the forward model and data to draw these conclusions, and did so independently for A and ϕ . The work of Seufert et al. (2011) performed a similar analysis, but instead divided Europa’s subsurface into many thin layers each with a different conductivity value. These authors determined the response ratio of the secondary (induced, ocean) field to the primary (inducing, Jovian) field at the surface, and used the response values of each layer to produce a more spatially refined conductivity model of the moon’s interior. Even with this substantially higher resolution of the subsurface conductivity, their constraints represent only a slight improvement upon the existing work of Zimmer et al. (2000). Similar to Zimmer et al. (2000), their sounding approach was rendered difficult by the lack of sufficiently diverse data coverage. Finally, the approach of Vance et al. (2021) analyzed the role of motional induction by currents in the moon’s ocean and found their contribution to range from fractions of a nT to nearly 20 nT depending on the presence, strength, and orientation of the fluid currents. They also independently re-analyzed the same Galileo flyby data as Zimmer et al. (2000) to find similar results, though extrapolating the values of A and ϕ directly to compositional breakdowns of the brines in Europa’s ocean.

To date, analyses to extract these inductive signals from Europa’s subsurface ocean have either performed only forward-modeling analysis or implemented a least-squares approach (e.g., Kivelson et al., 2002). Even state-of-the-art analyses using data from Juno, a magnetospheric survey mission that has performed a flyby of each Galilean moon in the extended phase, have only slightly increased the complexity of analogous inversions by using a weighted least-squares approach (Weber et al., 2022). The weighting scheme was motivated by the presence of well-constrained contaminating signals, an effort to not favor flybys with closer passes (i.e., with larger internal field magnitudes), and the desire to equally value multiple datasets with different measurement cadence. In addition, the analysis is often performed either for a single flyby or an limited set of flybys.

2 Science Questions

Therefore, we will partially deviate from the previous strategies and perform an inversion *only* on data from flybys corresponding to carefully selected intervals. These intervals are characterized (i) with a favorable inducing field that (theoretically) produces a strong, measurable oceanic response and (ii) without substantial disruptive signal from the plasma interaction. In this work, we will address the following science questions (SQ):

1. How can a model of Europa’s subsurface conductivity profile obtained from flyby data constrain the location, thickness, and composition of the subsurface ocean?
2. How are constraints on the recovery of Europa’s subsurface conductivity profile improved with a more sophisticated inversion algorithm (than Least-Squares)?
3. To what degree does a highly selective pre-screening routine applied to magnetometer data facilitate the characterization of Europa’s subsurface stratification?

Since there is already peer-reviewed literature available that addresses SQ1 alone (see Section 1.2), here we will focus on addressing SQ1&SQ2 or SQ1&SQ3 *in tandem*.

3 Behavior, Model, and Inversion of Europa’s Inductive Response

The inducing (primary) field perceived by Europa is equivalent to the equatorial projection of the Jovian magnetospheric field. A representative spectrum is included in Figure 1, sampled over 10 yr (adapted from Vance et al., 2021). In this figure and for the rest of the work, we operate in the satellite interaction system coordinates which admit the following basis. $\hat{\mathbf{x}}$ points in the direction of Europa’s orbit (and plasma flow), $\hat{\mathbf{z}}$ bears northward, and $\hat{\mathbf{y}}$ points toward Jupiter. To determine the ambient field at a particular location of Jupiter’s magnetosphere, we have implemented the Juno magnetospheric

field model determined by the spacecraft’s first 33 polar orbits through the system (Connerney et al., 2017; Connerney et al., 2018). This representation includes both a model for Jupiter’s internal field (through degree 18; see Connerney et al., 2022) and a model of the magnetic fields originating from currents in the magnetodisc plasma (Connerney et al., 2020). While the planetary synodic rotation period is the dominant oscillation (by an order of magnitude in power), other periods including Europa’s orbit, synodic harmonics, and beats of these (see Figure 1 and Zimmer et al., 2000; Vance et al., 2021). **We consider only induction signatures resultant from the synodic period in our analysis.**

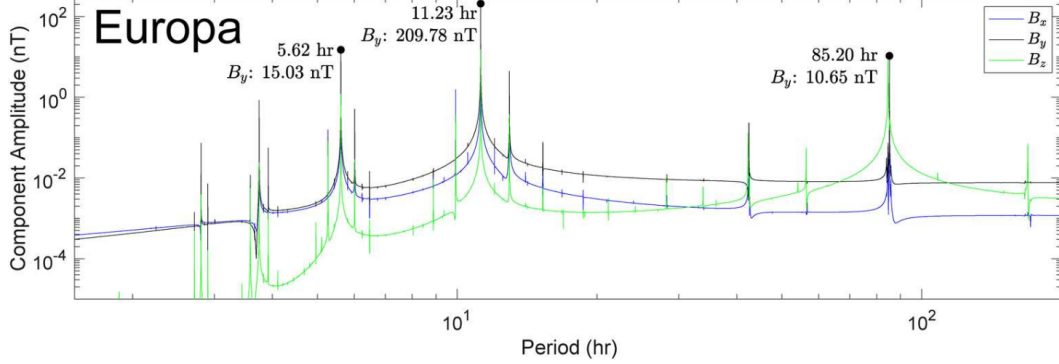


Figure 1. Spectrum of the magnetic field oscillations experienced by Europa over a window of 10 yr. The amplitude of the harmonic variations are given in nT (vertical axis) and the time is in hr (horizontal axis). The colors indicate the different components of $\mathbf{B}$. Adapted from Vance et al. (2021).

The forward model formalism, i.e., the inductive response of Europa with a specified subsurface conductivity (see Figure 2) is a solution to the diffusion equation:

$$\nabla^2 \mathbf{B} = \mu_0 \sigma \partial_t \mathbf{B} \quad ; \quad \text{if } \sigma \rightarrow 0 \quad \Rightarrow \quad \nabla^2 \mathbf{B} = 0 \quad . \quad (1)$$

As described briefly in Section 1.2, by assuming the inducing field is uniform over the moon’s body and imposing boundary conditions to ensure continuity and finiteness, the inductive response can be described as a dipole magnetic field with a moment pointing opposite to the inducing field (Zimmer et al., 2000; Rubin et al., 2015; Styczinski & Harnett, 2021). This also assumes that the conducting layer is not deep within the insulating ice, which is expected at Europa (Kivelson et al., 1999; Vance et al., 2018). This yields the induced magnetic dipole moment $\mathbf{M}_{\text{ind}}$ at Europa, centered at its core and given by (for external oscillation frequency ω)

$$\mathbf{M}_{\text{ind}} = -A e^{-i(\omega t - \phi)} \frac{2\pi R_E^3}{\mu_0} \mathbf{B}_{h,0} \quad , \quad (2)$$

where $\mathbf{B}_h$ is the horizontal component of the external Jovian field $\mathbf{B}_0$. For a perfect conductor, the value of A approaches unity and the phase lag ϕ vanishes. The negative sign leading equation (2) indicates that the field directly opposes the primary magnetospheric inducing field. A more rigorous treatment of the inductive response would include e.g., radial temperature gradients that map into the conductivity profile, motional induction by fluid currents, and an anisotropic structure of the conducting layer; however, these effects have all been shown to be weak, each contributing $\sim 1\%$ at most to the modified field above Europa’s surface (e.g., Schilling et al., 2007; Seufert et al., 2011; Vance et al., 2021; Styczinski & Harnett, 2021). A finite conductivity of the ocean layer results in modified values for A and ϕ . **The field at position $\mathbf{r}$ is $\mathbf{B}_{\text{ind}} = \frac{\mu_0}{4\pi} (3(\mathbf{r} \cdot \mathbf{M}_{\text{ind}})\mathbf{r} - r^2 \mathbf{M}_{\text{ind}}) / r^5$.**

Rendering further confidence in our approach, this expression has been applied extensively in works that study the magnetosphere-atmosphere or magnetosphere-subsurface interaction of Europa (e.g., Rubin et al., 2015; Harris et al., 2021; Addison et al., 2021,

2024; Haynes et al., 2023) and other icy Galilean moons with putative subsurface oceans (e.g., M. Kivelson et al., 1996; Stahl et al., 2023; Haynes, Tippers, et al., 2025; Haynes, Simon, & Liuzzo, 2025).

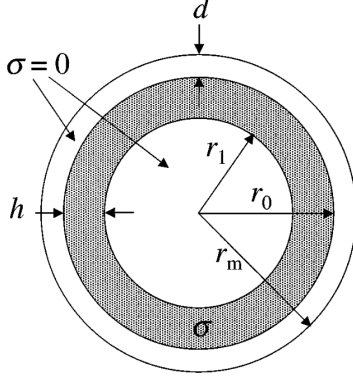


Figure 2. Radially symmetric structure assumed for Europa's subsurface. Adapted from (Zimmer et al., 2000).

Following precedent in the peer-reviewed literature on probing Europa's subsurface structure via induction, we assume that the moon's underlying layers are described as follows. Each layer of the moon is spherically symmetric. The subsurface is distributed into a core/mantle with conductivity $\sigma = 0$, an ocean layer of conductivity σ_o with inner radius r_1 and outer radius r_0 , and an insulating crust with $\sigma = 0$. This structure is illustrated in Figure 2. Though not directly corresponding to a subsurface conductivity profile, the parameters A and ϕ can be expressed in terms of σ_o , r_1 , r_0 . Following Zimmer et al. (2000) and Parkinson (1983), using Bessel functions of the first kind J_m of order m , it can be shown that

$$Ae^{i\phi} = \left(\frac{r_0}{R_E}\right)^3 \frac{C J_{\frac{5}{2}}(r_0 k) - J_{-\frac{5}{2}}(r_0 k)}{C J_{\frac{1}{2}}(r_0 k) - J_{-\frac{1}{2}}(r_0 k)} \quad , \quad (3)$$

where

$$C \equiv \frac{r_1 k J_{-\frac{5}{2}}(r_1 k)}{3 J_{\frac{3}{2}}(r_1 k) - r_1 k J_{\frac{1}{2}}(r_1 k)} \quad . \quad (4)$$

In equations (3) and (4), the complex wavenumber parameter

$$k = (1 - i) \sqrt{\frac{\mu_0 \sigma_o \omega}{2}} = (1 - i) \sqrt{\frac{\pi \mu_0 \sigma_o}{T_s}} \quad (5)$$

is dependent on the conductivity and driving frequency. Throughout this work, we assume the magnetic permeability μ of Europa to be its value in a vacuum μ_0 (Vance et al., 2021).

We will retain the assumption of a uniform ocean conductivity in order to ensure the model inversion is tractable, sacrificing a higher order correction to the internal conductivity profile of the moon (see also Seufert et al., 2011; Vance et al., 2021). Our strategy to address the **SQs** (see Section 2) is outlined as follows:

1. Identify appropriate segments of Europa flybys from Galileo (Kivelson et al., 2009) and Juno (Addison et al., 2023, 2024).
2. Perform a downsampling routine where appropriate to ensure that the sampling rate is equal across all datasets, following the approach of Weber et al. (2022).
3. Detrend the data by subtracting the magnetospheric background (primary) field from the observed signatures. If plasma simulation output exists and is available to the author, as is the case for the Juno flyby of Europa (Addison et al., 2024), incorporate the plasma currents into the background subtraction routine.
4. Perform an inversion for the data to constrain the outer radius r_0 , inner radius r_1 , ocean conductivity σ_o and phase shift ϕ . Note that, at least preliminarily, we assume the conductivity to be a constant value within Europa's ocean.
5. Start with Least-Squares (L-S) and weighted L-S, advancing to more sophisticated or appropriate techniques after results from the literature have been replicated (Zimmer et al., 2000; Kivelson et al., 2000). See also Section 6.

Together, steps 1, 3, and 4 address **SQ1&SQ3**, while steps 4 and 5 address **SQ1&SQ2**.

4 Synthetic Forward Model Output

In the following, we calculate and display synthetic forward model output (i.e., $\mathbf{B}_{\text{ind}}$) for several simplified models that are representative of our problem and approach.

4.1 Idealized Case: Perfect Conductor

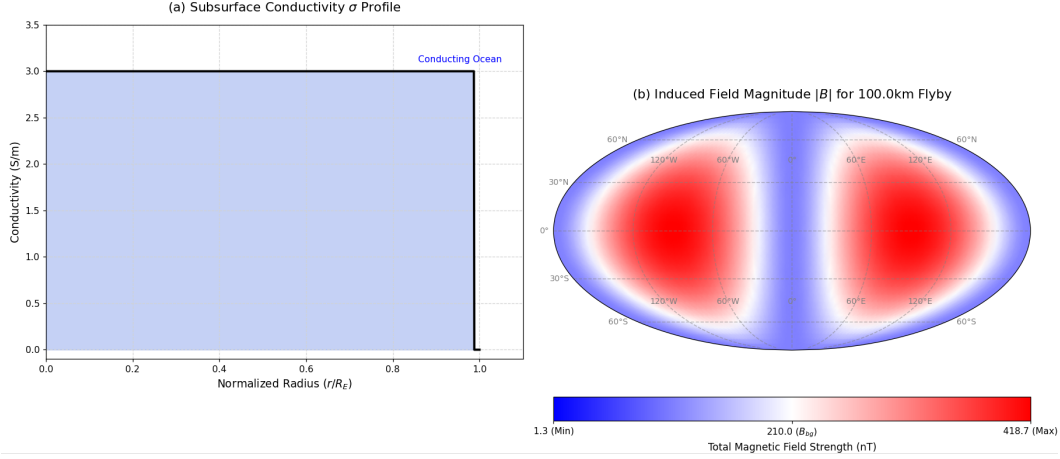


Figure 3. Subsurface conductivity profile (panel (a)) and associated induction signature $|\mathbf{B}|$ (panel (b)) projected onto Europa’s surface. The moon is assumed to be a perfectly conducting sphere of radius R_E . Note that the absolute value of conductivity is actually ∞ , but is set to a finite value for visualization purposes. The inducing field is assumed to be in the $-y$ direction ($-x$ direction points out of the screen).

In the extreme case, i.e., that of a perfectly conducting moon, a time-varying magnetic field does not penetrate the surface of a perfect conductor because it is perfectly opposed along the axis of $\mathbf{B}_h$. However, even though the time-varying component of the Jovian field is expelled from the moon’s volume, the stronger southward component causes magnetospheric field lines to still thread Europa in the case it is perfectly conducting (Zimmer et al., 2000; Seufert et al., 2011). This case is depicted for Europa in Figure 3, where we have assumed $\sigma \rightarrow \infty$ and $\phi = 0^\circ$. The magnetic perturbations at the surface demonstrate how for a perfect conductor, the field magnitude approaches 0 nT at the two locations where the induced moment is aligned with the inducing field. The conductivity profile in panel 3(a) is merely fiducial, since a finite conductivity value (even throughout the whole body of the moon) would not yield a perfect conductor response, but would approach one as the conductivity was increased above ~ 1 S/m. This also demonstrates the general signature observable near Europa’s surface for an induced field being present.

4.2 Simplified Realistic Conductivity Model: 3-Layer Approximation

Following the discretization of Europa’s subsurface described in Section 3, we now demonstrate the effects that a finite ocean thickness and conductivity have on the induction signal at the surface. To do so, we depict three conductivity models for Europa’s subsurface with the associated inductive response. For each model, we assume that the outer radius of the ocean is $r_0 = R_E - 20$ km. Then, by varying both the assumed ocean thickness of the model $r_0 - r_1$ and the conductivity σ_o , we can depict how Europa’s inductive response changes. In all cases, we assume the primary field to be given by $\mathbf{B}_0 = \mathbf{B}_h \equiv -210$ nT for simplicity (though this is a realistic choice for $\mathbf{B}_h$ alone; see, e.g., Kivelson et al., 2009; Addison et al., 2021).

Figure 4 depicts Europa’s subsurface conductivity profile and the magnetic field magnitude $|\mathbf{B}|$ at Europa’s surface. Because the fields add linearly, $\mathbf{B} = \mathbf{B}_0 + \mathbf{B}_{\text{ind}}$. To

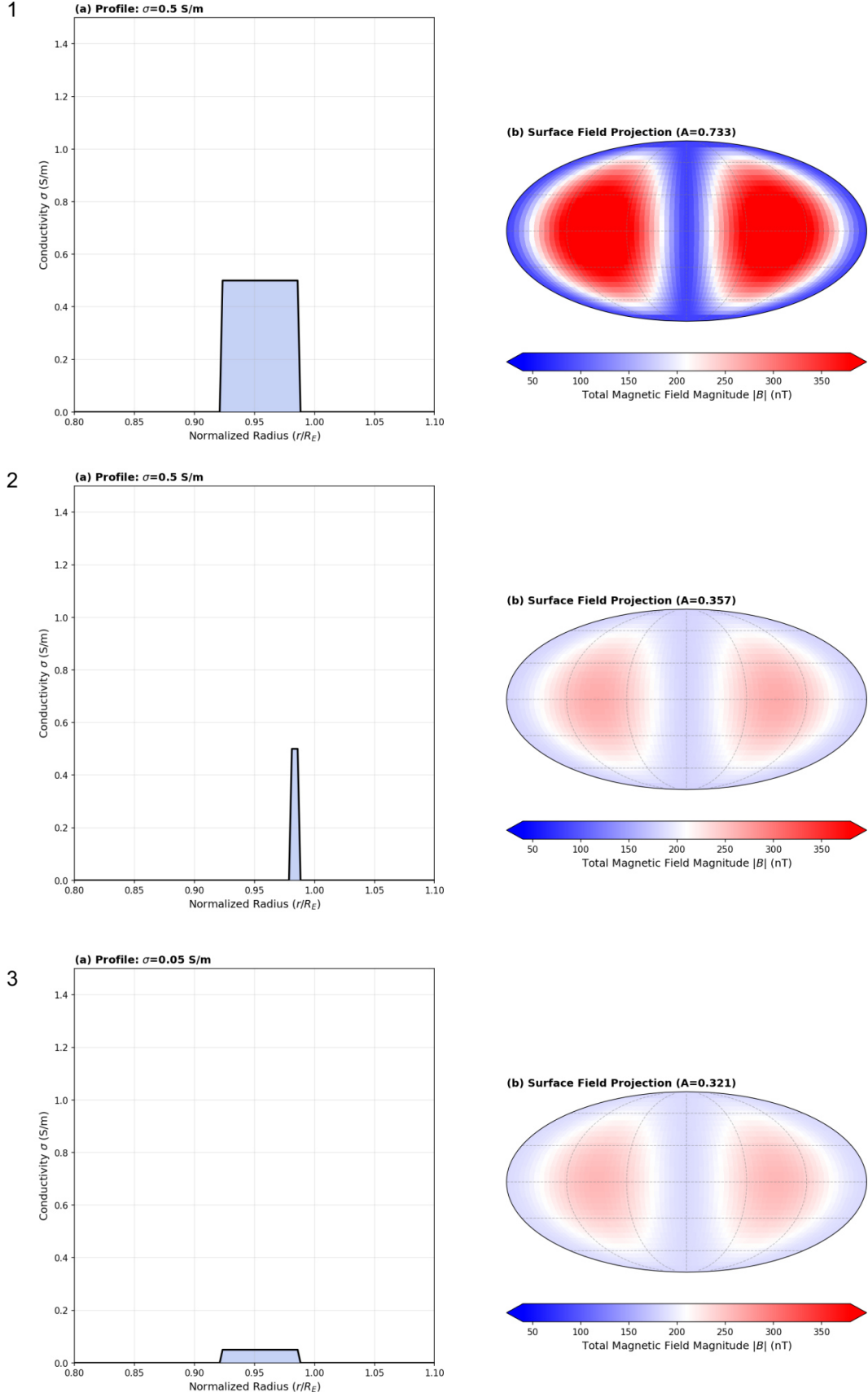


Figure 4. Subsurface conductivity profile (panels (a)s) and associated induction signature $|B|$ (panel (b)s) projected onto Europa's surface. The different rows depict different assumed models, to indicate how the induction signature from Europa's ocean is sensitive to the parameters in our model. For all rows, the inducing field is assumed to be in the $-y$ direction ($-x$ direction is coming out of the screen from this vantage) and the crust thickness is 20 km. In row 1, the conductivity assumed for the ocean is $\sigma_o = 0.5$ S/m, and the resultant amplitude from equation 3 is given by $A = 0.733$. The ocean thickness is assumed to be 100 km. In the middle row (2), the ocean thickness is reduced to be 10 km, using the same conductivity of $\sigma_o = 0.5$ S/m. This yields a value of $A = 0.357$ for the amplitude. In row 3, the conductivity assumed for the ocean is $\sigma_o = 0.05$ S/m, in contrast to rows 1 and 2. The amplitude yielded from equation 3 using these parameters is given by $A = 0.321$.

generate these, we assume a negligible phase lag. Accordingly, the orientation of the poles of $|\mathbf{B}_{\text{ind}}|$ nearly exactly oppose the external inducing field $\mathbf{B}_0$, which we set to $\mathbf{B}_0 = \mathbf{B}_h = 210 \text{ nT}$ for simplicity (i.e., we neglect the non-inducing, southward component of the Jovian magnetospheric field). In panel 4(1a) in the upper-left, the subsurface conductivity profile for an assumed value of $\sigma_o = 0.5 \text{ S/m}$ is displayed with a thickness of 100 km and a depth of 20 km. Then, panel 4(1b) illustrates the magnitude of the net magnetic field across Europa's surface. The second and third rows are arranged in an analogous manner, and use the same assumptions for Europa's crustal thickness. However, the middle row of Figure 4 shows the conductivity profile and induced field when the ocean thickness is drastically reduced to only 10 km, which we note is thinner than most current-standing constraints (Vance et al., 2021). Similarly, the bottom row preserves the same thickness assumption for the ocean of 100 km, but assumes a reduced conductivity of only 50 mS/m, also at the lower edge of available constraints (Zimmer et al., 2000; Seufert et al., 2011). In both rows 2 and 3 of Figure 3, the resultant induction signature has a markedly weaker amplitude ($A = 0.357$ and $A = 0.321$, respectively) than that of an ocean 100 km thick with a conductivity of 500 mS/m. This provides some context for the frequently taken assumption of a perfect conductor for Europa's inductive behavior in studies that don't seek to constrain the ocean (but rather plasma effects; see, e.g., Haynes et al., 2023): an ocean only 100 km thick and with a conductivity of 500 mS/m is already 73% the amplitude of a perfect response.

5 Observational Noise

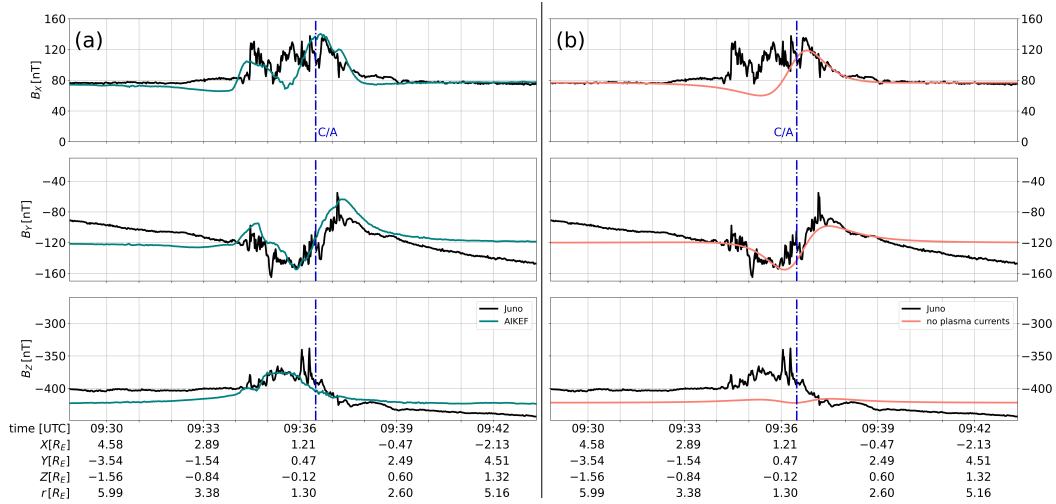


Figure 5. Plots of the vector magnetic field components B_x (top), B_y (middle), and B_z (bottom) observed by the Juno spacecraft on its sole flyby of Europa. Panel (a) compares the Juno magnetometer measurements (black) to AIKEF simulation output for Europa's interaction (shown in teal), and panel (b) shows a comparison between the data and a model of the moon's induced field signature (shown in pink) assuming a perfect conductor, i.e., $A = 1$, $\phi = 0$ (see Addison et al., 2024). The blue line in each panel marks the time of Juno's closest approach (C/A). The right panel shows noisy data (black) versus noiseless data (pink).

In Figure 5, we demonstrate the true nature of magnetic field observations near Europa. The exact behavior of the fluctuations in the magnetospheric field near Europa is poorly-constrained, erratic, and multi-faceted, stemming from a multitude of effects with varying degrees of theoretical understanding. Therefore, rather than making an attempt at analytically replicating these different sources of noise, we present a direct observation of these signatures from Juno's fluxgate magnetometer (Connerney et al., 2017). This data was collected as the spacecraft flew by Europa on September 09, 2022. The time and position coordinates are shown along the horizontal axis. It is contrasted against a

model of Europa’s induced field signature assuming a perfect conductor response (pink curve) as well as simulation output from the AIKEF hybrid model (kinetic ions, fluid electron treatment; see Müller et al., 2011) published by Addison et al. (2024). It should be noted that the background magnetospheric field $\mathbf{B}_0$ is included in each component, and is the cause of the apparent variation in the background field along the flyby (most visible in the B_y component). This was caused primarily by the rapid motion of the Juno spacecraft, *not* the rotation of the Jovian field in Europa’s frame, as the flyby occurred along a tiny fraction of T_s (see timestamp labels). Hence, the pink curve represents a superposition of Europa’s perfect conductor induction signature and the Jovian background field. Visual inspection indicates that there are marked differences between the induced field signature and the measured magnetic field near Europa.

At Europa, upstream magnetospheric plasma densities range from $(40\text{--}200)\text{ cm}^{-3}$ (Bagenal & Delamere, 2011; Roth et al., 2014). When the moon is near the magnetic equator, the inducing component of the primary field is weakest, and the plasma interaction is the strongest. In contrast, when near its maximum distance above or below the magnetic equator, the horizontal component $\mathbf{B}_h$ is at its peak strength and the plasma interaction is weakest (e.g., Haynes et al., 2023). Perturbations stemming from the plasma interaction are often on the order of 20–50% the background field values. That is, the magnetic field signatures from the plasma interaction may have a similar magnitude as $|\mathbf{B}_h|$. Thus, even for a perfectly conducting moon (Section 4.1), the induced field is still of the same order of magnitude as the plasma interaction near the surface. Since the dipolar form of the induced field falls off with $\propto 1/r^3$, it is often infeasible to extract meaningful induction signatures from magnetometer readings captured at high altitudes for a flyby near the magnetospheric equator. Therefore, ideal flybys or flyby segments for the recovery of an induction signal are far more likely to occur when the moon is far above or below the Jovian magnetic equator.

Even the sophisticated hybrid simulation of Addison et al. (2024) is not able to capture some structures in the plasma interaction as well as rapid fluctuations associated with transient signatures. Since we have described the plasma interaction in detail (see Section 1.1), we now name a few other forms of noise that contaminate the magnetometer readings captured near the moon by spacecraft such as Juno. In addition to magnetic signatures, the perturbations generated by the strong plasma interaction (Saur et al., 1999) and ion production at the moon drive the production of a plethora of plasma waves (Kivelson et al., 1996; Simon et al., 2021). Such waves include the Alfvén wave mode, compressional fast and slow modes, ion cyclotron modes, (right-hand polarized) ULF modes, whistler wave modes, and more (e.g., Neubauer, 1980, 1998; Volwerk et al., 2001; Simon et al., 2011; Menietti et al., 2012; Shprits et al., 2018; Saur, 2021). Each of these waves produces fluctuations (at least in individual components, if not the magnitude) in $\mathbf{B}$ that are detectable in the vicinity of the moon (see high-frequency oscillations in the black curves of Figure 5). Moreover, the plasma torus and magnetospheric conditions upstream of Europa may not always be in steady-state due to temperature and density fluctuations (Bagenal & Dols, 2020) or magnetospheric interchange and global cycling (Vasyliunas, n.d.).

6 Summary and Outlook on Inversion Strategies

In this work, we have provided the background, motivation and outlined our approach for addressing the **SQs** listed in Section 2. In brief, we seek to perform an inversion of magnetometer data collected by the Galileo and Juno spacecraft near Europa to constrain the subsurface conductivity profile. This corresponds directly to a depth, thickness, and composition of Europa’s putative subsurface liquid water ocean, constraints upon which remain actively sought after. To do so, we will carry out the steps detailed in Section 3. To begin, we assume a coarse, 3-layer model for Europa’s subsurface that has extensive precedent in the peer-reviewed literature (see Section 1.2); the ocean is set to a constant conductivity, surrounded by a thin $\sim \mathcal{O}(10\text{ km})$, insulating icy crust while

surrounding itself an insulating mantle and core. This approach seeks to invert observed induction signatures to determine the depth, thickness, and conductivity of the ocean layer. The induction signal also depends on a phase lag corresponding to the nonzero time it takes the electromagnetic signal to propagate through a solid medium, which is partially dependent on the other three parameters. Therefore, we can provide understanding for Europa’s subsurface stratification as well as the composition of its subsurface ocean.

Our strategy for the inversion itself will rely on the “overdetermined” nature of our model setup: since we have only 3-4 free parameters describing our conductivity model (depending on how we decide to constrain ϕ), and *many* vector readings near Europa along different flybys, the shape of the system will correspond to an extremely tall matrix. We could increase the “squareness” (i.e., how determined our system is) by introducing many small layers each with a different conductivity value for the ocean, similar to (Seufert et al., 2011) and Vance et al. (2021). However, for several reasons, our formulation of the problem is far less overdetermined than it appears mathematically (S. Naif, 2026, personal communication)¹. The spacecraft sampling the magnetic field signatures are at constantly changing altitudes, some distance above the surface. Therefore, any small anisotropies in the structure of the induced field will be sampled uniquely at different positions, introducing systematic error to our data. Furthermore, signatures in the magnetic field corresponding to plasma effects are constantly prevalent in the readings as well (see Figure 5). In addition, there are several effects that would produce a slightly non-dipolar or anisotropic signature that we have neglected in our approach (see Section 3). Therefore, we proceed with our 3-layer approximation with a constant conductivity.

As described in Section 2, we intend to begin with a Least-Squares approach for the inversion to replicate results in the literature (Zimmer et al., 2000), then advancing to more appropriate and robust techniques once we have done so. Our inversion strategy will be motivated by a mathematically overdetermined, nonlinear system with substantial and constantly prevalent noise. So far, we can see two possible approaches: on one hand, we can envision performing a regularization and truncation of nonlinear L-S to create a well enough constrained model from the data. Because we have a system where there are substantially more rows in our forward operator than columns, we can use truncation to favor the “most important” information to consider. On the other hand, an iterative method such as nonlinear Conjugate Gradients may be more appropriate and successful. Similarly, we can select a truncation threshold (i.e., a number of iterations of Conjugate Gradients) that corresponds to regularizing our problem. However, it remains to be understood how the nonlinear implementation works and whether it is tractable for a final project. The nonlinear nature of our operator (see Section 3) may necessitate an adjusted approach that is tailored to handle such relationships.

¹ While the author disagrees with the notion of citing a personal communication, we are merely attempting to jog the memory of the reader regarding a past conversation.

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